

CodeAlpha IoT Internship

Task 2: Sensor-Based Simulation Report

Project Title: Automatic Night Light System **Simulation Platform:** Tinkercad Circuits **Microcontroller:** Arduino Uno R3
Domain: Internet of Things (IoT) **Input Sensor:** LDR (Photoresistor) **Output Actuator:** LED Indicator

1. EXECUTIVE OVERVIEW

This report documents the design, implementation, and simulation of an **Automatic Dark-Detector Night Light** built as part of **Task 2 (Sensor-Based Simulation)** for the CodeAlpha IoT Internship program.

The objective of this task is to design a sensor-driven embedded system in a simulation environment without physical hardware constraints. The system monitors ambient light levels via a Photoresistor (Light Dependent Resistor - LDR) and automatically activates an LED light source when light intensity falls below a predetermined safety threshold.

2. HARDWARE COMPONENTS & SPECIFICATIONS

COMPONENT	QUANTITY	SPECIFICATIONS / RATING	ROLE IN CIRCUIT
Arduino Uno R3	1	ATmega328P Microcontroller (5V)	Central processing unit reading sensor data and controlling output pin state.
Photoresistor (LDR)	1	Light-Dependent Resistor (1kΩ – 100kΩ)	Primary analog sensor detecting ambient light brightness.
Fixed Resistor	1	10 kΩ (Brown-Black-Orange)	Pull-down resistor paired with LDR to form a standard Voltage Divider network.
LED (Red)	1	Standard 5mm Red LED	Visual actuator acting as the automated night light.
Current Limiting Resistor	1	220 Ω (Red-Red-Brown)	Prevents excessive current through the LED to prevent overcurrent damage.
Breadboard & Jumpers	1	Small Breadboard & 5V/GND Wires	Provides structural connections between components and Arduino headers.

3. CIRCUIT ARCHITECTURE & PIN CONNECTIONS

The system comprises two core sub-circuits: the **LDR Sensor Input Interface** and the **LED Actuator Output Interface**.

A. Voltage Divider Network (LDR Input)

Because an LDR varies its internal resistance based on light exposure, the Arduino cannot read resistance directly. To convert variable resistance into a readable voltage signal (V_{out}), the LDR and a 10 k Ω resistor are connected in series:

- **LDR Leg 1:** Connected to Arduino **5V Power Rail**.
- **LDR Leg 2 (Junction Point):** Connected simultaneously to:
 1. Arduino Analog Input Pin **A0** (Voltage sensing).
 2. One leg of the **10 k Ω pull-down resistor**.
- **Other leg of 10 k Ω Resistor:** Connected to Arduino **GND Rail**.

The output voltage is given by: $V_{out} = 5V \times (R_{10k} / (R_{LDR} + R_{10k}))$

B. LED Output Interface

- **Arduino Digital Pin 13:** Connects to one leg of the **220 Ω resistor**.
- **Resistor Leg 2:** Connects to the **LED Anode** (longer positive leg).
- **LED Cathode:** Connects to the Arduino **GND Rail**.

4. SOURCE CODE (`MAIN.INO`)

```
// =====  
// CodeAlpha IoT Internship – Task 2: Sensor-Based Simulation  
// Project: Automatic Dark-Detector Night Light System  
// Platform: Tinkercad / Arduino Uno  
// =====  
  
// Pin Configuration  
const int ldrPin = A0;    // Analog pin connected to Voltage Divider junction  
const int ledPin = 13;    // Digital pin connected to LED Anode  
  
// Variable Declarations  
int lightLevel = 0;       // Holds raw 10-bit analog reading (0 to 1023)  
int threshold = 500;      // Calibrated ambient light threshold  
  
void setup() {  
    pinMode(ledPin, OUTPUT); // Configure digital pin 13 as digital output  
    Serial.begin(9600);      // Initialize serial communications at 9600 baud  
    Serial.println("System Initialized. Starting Light Monitoring...");  
}  
  
void loop() {  
    // Read analog voltage from LDR voltage divider circuit  
    lightLevel = analogRead(ldrPin);  
  
    // Output real-time reading to Serial Monitor for system calibration  
    Serial.print("Current Light Sensor Level: ");  
    Serial.println(lightLevel);  
  
    // Decision Logic: Switch light ON in darkness, OFF in brightness
```

```
if (lightLevel < threshold) {  
    digitalWrite(ledPin, HIGH); // Turn ON LED  
} else {  
    digitalWrite(ledPin, LOW);   // Turn OFF LED  
}  
  
delay(100); // 100ms sampling delay for sensor measurement stability  
}
```

5. OPERATIONAL LOGIC & TESTING RESULTS

The system execution cycle operates continuously inside the micro-controller loop:

CONDITION	LDR RESISTANCE	ANALOG READ (A0)	LOGIC DECISION	LED STATUS
BRIGHT LIGHT	Low (~1 kΩ or less)	High (> 500)	$lightLevel \geq threshold$	OFF (LOW)
DARKNESS / SHADOW	High (~100 kΩ or more)	Low (< 500)	$lightLevel < threshold$	ON (HIGH)

Verification in Tinkercad Simulator:

- Simulation Start:** Upon pressing *Start Simulation*, the Arduino initializes pin modes and starts transmitting readings to the Serial Monitor at 9600 baud.
- Daytime Simulation:** Adjusting the Tinkercad LDR slider to full light yields readings around **700–900**. Pin 13 remains LOW; LED is OFF.
- Nighttime Simulation:** Dragging the slider towards dark drops readings below the threshold value (e.g., **< 300**). Pin 13 immediately switches to HIGH, causing the LED to illuminate in the simulator window.

6. CONCLUSION & REAL-WORLD APPLICATIONS

The **Task 2 Sensor-Based Simulation** successfully demonstrates automated environmental response using low-cost electronic components. The circuit eliminated short circuit risks, properly implemented a sensor voltage divider network, and verified trigger thresholds digitally.

Real-World IoT Applications:

- Smart Street Lighting:** Municipal streetlamps automatically toggling based on sunset/sunrise to conserve electrical grid energy.
- Home Automation:** Indoor ambient accent lamps automatically turning on during nighttime.
- Solar Energy Harvesting:** Solar power storage management systems toggling between charge and discharge modes based on solar irradiance levels.